

An empirical example: Four terminals and two areas

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```
# Four taxa and two areas
```

```
## Libraries
```

```
library(blepd)
```

```
## Loading required package: ape
```

```
## Loading required package: picante
```

```
## Loading required package: vegan
```

```
## Loading required package: permute
```

```
##
```

```
## Attaching package: 'permute'
```

```
## The following object is masked from 'package:devtools':
```

```
##
```

```
##      check
```

```
## Loading required package: lattice
```

```
## This is vegan 2.6-6.1
```

```
## Loading required package: nlme
```

```
library(ggtree)
```

```
## ggtree v3.10.1 For help: https://yulab-smu.top/treedata-book/
```

```
##
```

```
## If you use the ggtree package suite in published research, please cite
```

```
## the appropriate paper(s):
```

```
##
```

```
## Guangchuang Yu, David Smith, Huachen Zhu, Yi Guan, Tommy Tsan-Yuk Lam.
```

```
## ggtree: an R package for visualization and annotation of phylogenetic
```

```
## trees with their covariates and other associated data. Methods in
```

```
## Ecology and Evolution. 2017, 8(1):28-36. doi:10.1111/2041-210X.12628
```

```
##
```

```
## Guangchuang Yu. Using ggtree to visualize data on tree-like structures.
```

```
## Current Protocols in Bioinformatics. 2020, 69:e96. doi:10.1002/cpbi.96
```

```
##
```

```
## LG Wang, TTY Lam, S Xu, Z Dai, L Zhou, T Feng, P Guo, CW Dunn, BR
```

```
## Jones, T Bradley, H Zhu, Y Guan, Y Jiang, G Yu. treeio: an R package
```

```
## for phylogenetic tree input and output with richly annotated and
```

```
## associated data. Molecular Biology and Evolution. 2020, 37(2):599-603.
```

```
## doi: 10.1093/molbev/msz240
##
## Attaching package: 'ggtree'
## The following object is masked from 'package:nlme':
##
##      collapse
## The following object is masked from 'package:ape':
##
##      rotate
library(ggplot2)

## Data

# Call the 4 terminals example

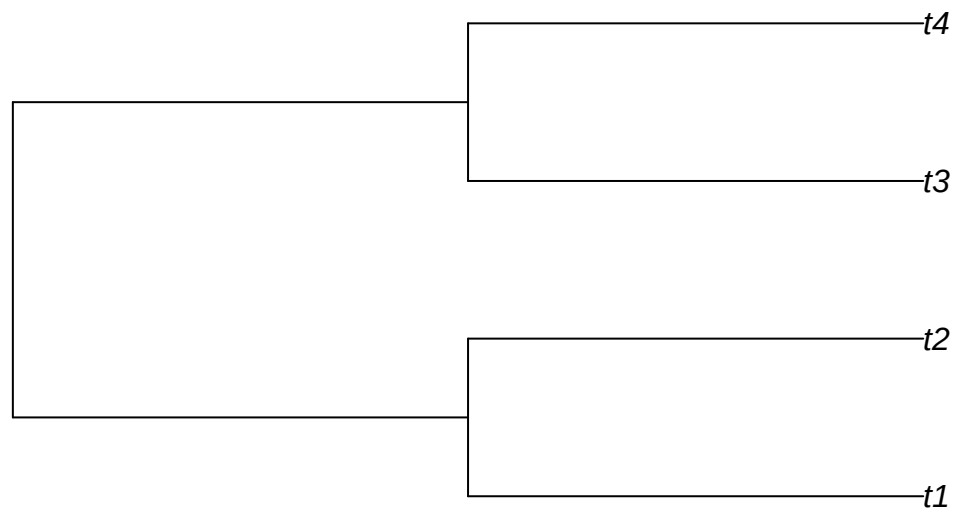
##data(package = "blepd")

## trees

data(tree)

#str(tree)

plot(tree,use.edge.length =T)
```



```

initialTree <- tree

## distributions

data(distribution)

##str(distribution)

dist4taxa <- distribution

## distribution to XY to be able to plot the data, otherwise, skip the next step

distXY <- matrix2XY(dist4taxa)

distXY

## Terminal Area
## 1      t1    A1
## 2      t3    A1
## 3      t2    A2
## 4      t4    A2

## plotting

## the tree

```

```

plotTree <- ggtree(initialTree, ladderize=TRUE,
                   color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=4.5, color="black") +
  theme_tree2() +
  ggtitle("Four terminals, all branches are == 1")

##
print(plotTree)

```

Four terminals, all branches are == 1



```

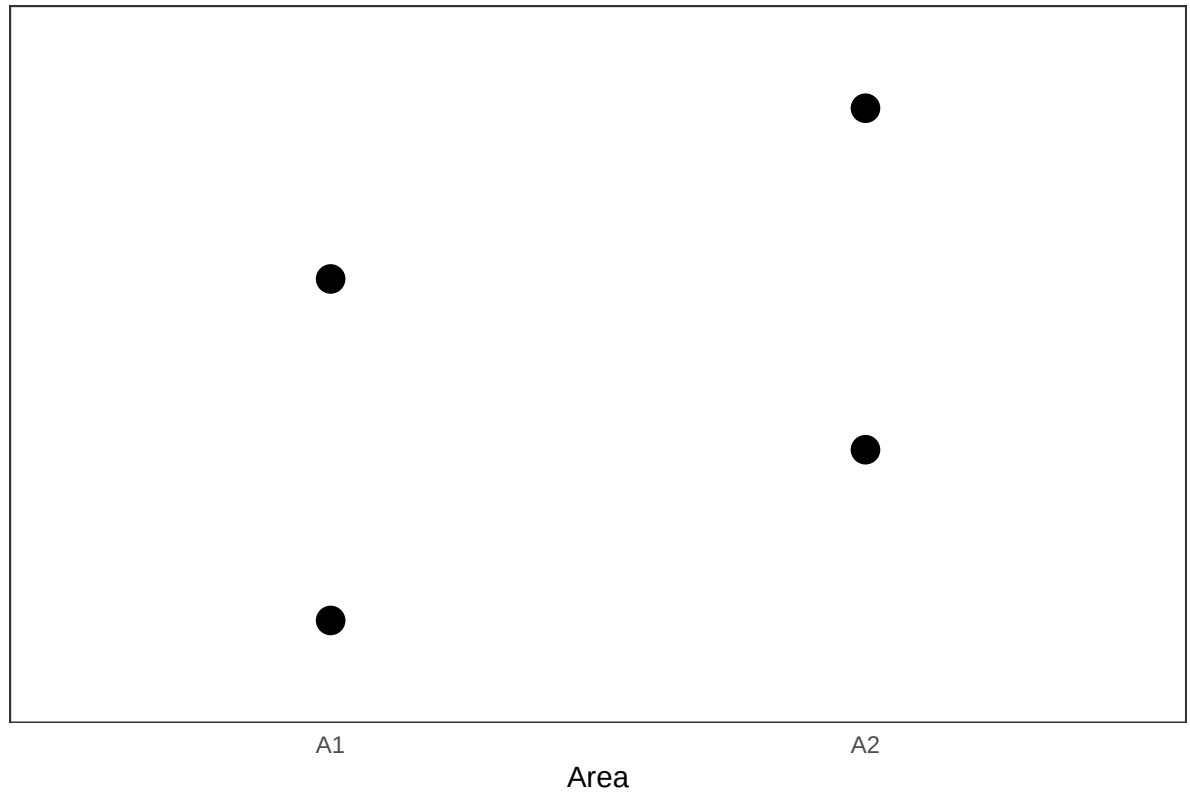
## the distribution

plotDistrib <- ggplot(data=distXY,
                     aes(x= Area, y= Terminal , size =150)) +
  geom_point() +
  theme_bw() +
  theme( axis.text.y = element_blank(),
        panel.grid.major = element_blank(), # Remove major gridlines
        panel.grid.minor = element_blank(), # Remove minor gridlines
        axis.line = element_line(colour = NA_character_), # Remove axis lines
        axis.ticks = element_blank(), # Remove axis ticks
        legend.position = "none"
  ) +
  labs(title = "Distributions",
       y = "",

```

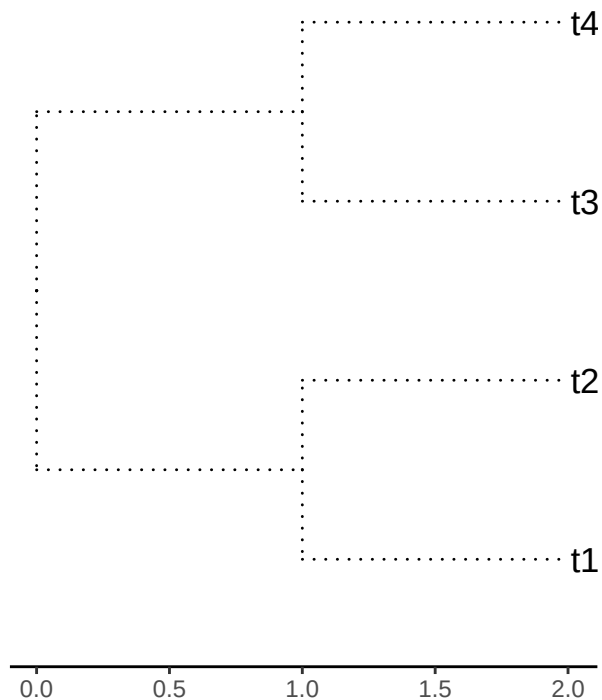
```
x = "Area")  
  
##  
print(plotDistrib)
```

Distributions

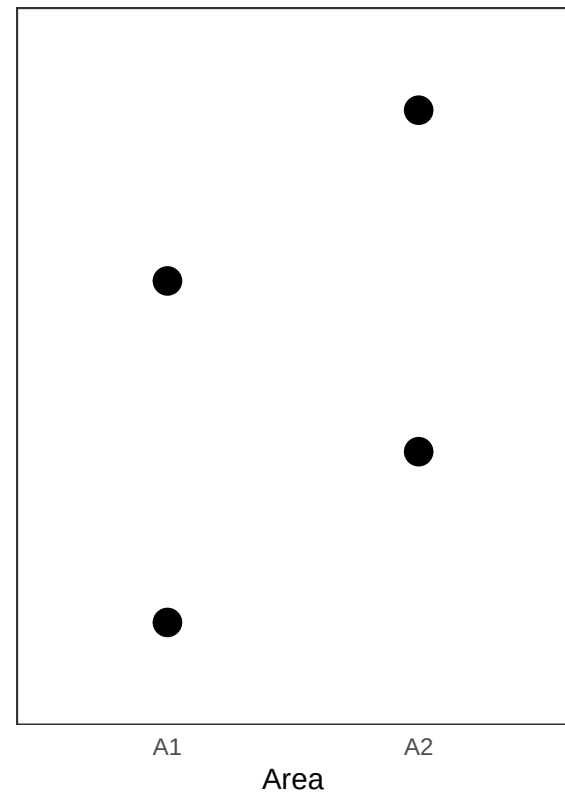


```
cowplot::plot_grid(plotTree, plotDistrib, ncol=2)
```

Four terminals, all branches are == 1



Distributions



```
#We check whether names in both objects: initialTree and dist4taxa, are the same.
```

```
##all(colnames(dist4taxa) == initialTree$tip.label)
```

```
all(colnames(dist4taxa) %in% initialTree$tip.label)
```

```
## [1] TRUE
```

```
#We report the branch length, and calculate the PD values.
```

```
initialTree$edge.length
```

```
## [1] 1 1 1 1 1 1
```

```
initialPD <- PDindex(tree=initialTree, distribution = dist4taxa)
```

```
initialPD
```

```
## [1] 4 4
```

```
#As expected there is a tie between both areas.
```

```
#Now, we can see the impact of swapping the branch lengths; we will use the three
#available models to swap: "simpleswap", "allswap", "uniform", and we will swap the
#three classes of branches: "terminals", "internals", "all".
```

```
##?swapBL
```

```
for( modelo in c("simplswap","allswap","uniform") ){  
  for( rama in c("terminals","internals","all") ){  
    ## cat( "\n\t Branchs swapped=\t",rama,".\t",sep="" )  
    val <- swapBL(tree=initialTree,  
                  distribution = dist4taxa,  
                  model = modelo,  
                  branch = rama  
                  )  
    printBlepd(val, compact=TRUE)  
  
    cat( "\n\n" )  
  }  
}
```

```
## model to test simplswap reps 100  
## AreaSelected Freq  
## 1          A1A2  100  
##  
##  
## model to test simplswap reps 100  
## AreaSelected Freq  
## 1          A1A2  100  
##  
##  
## model to test simplswap reps 100  
## AreaSelected Freq  
## 1          A1A2  100  
##  
##  
## model to test allswap reps 100  
## AreaSelected Freq  
## 1          A1A2  100  
##  
##  
## model to test allswap reps 100  
## AreaSelected Freq  
## 1          A1A2  100  
##  
##  
## model to test allswap reps 100  
## AreaSelected Freq  
## 1          A1A2  100  
##  
##  
## model to test uniform reps 100
```

```

##   AreaSelected Freq
## 1           A1A2 100
##
##
## model to test uniform reps 100
##   AreaSelected Freq
## 1           A1A2 100
##
##
## model to test uniform reps 100
##   AreaSelected Freq
## 1           A1A2 100

val$bestModifiedArea

##   AreaSelected Freq
## 1           A1A2 100

t(val$bestModifiedArea)

##           [,1]
## AreaSelected "A1A2"
## Freq         "100"

data.frame(t(val$bestModifiedArea))

##           t.val.bestModifiedArea.
## AreaSelected           A1A2
## Freq           100

b <- data.frame(t(val$bestModifiedArea))

colnames(b) <- b[1,]
b <- b[2,]
class(b)

## [1] "character"

str(b)

## chr "100"

print(b)

## [1] "100"
##

#As expected, as all braches are EQUAL nothing happens

#Now, let us see if the results could be related to the branch length (or not).

## we could use a single command (*evalBranch*)

```



```

##?evalBranch

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")

## Loading required package: maps
##
## Attaching package: 'phytools'
## The following object is masked from 'package:vegan':
##
##     scores
## let's see the structure of the output

##str(AllTerminals)

## Now we can print a more friendly output

printEvalBranch(AllTerminals, compact=FALSE)

## data frame with 0 columns and 0 rows
## Now the internals

AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")

printEvalBranch(AllInternals, compact=FALSE)

## data frame with 0 columns and 0 rows
##now with a longer branch

initialTree <- tree

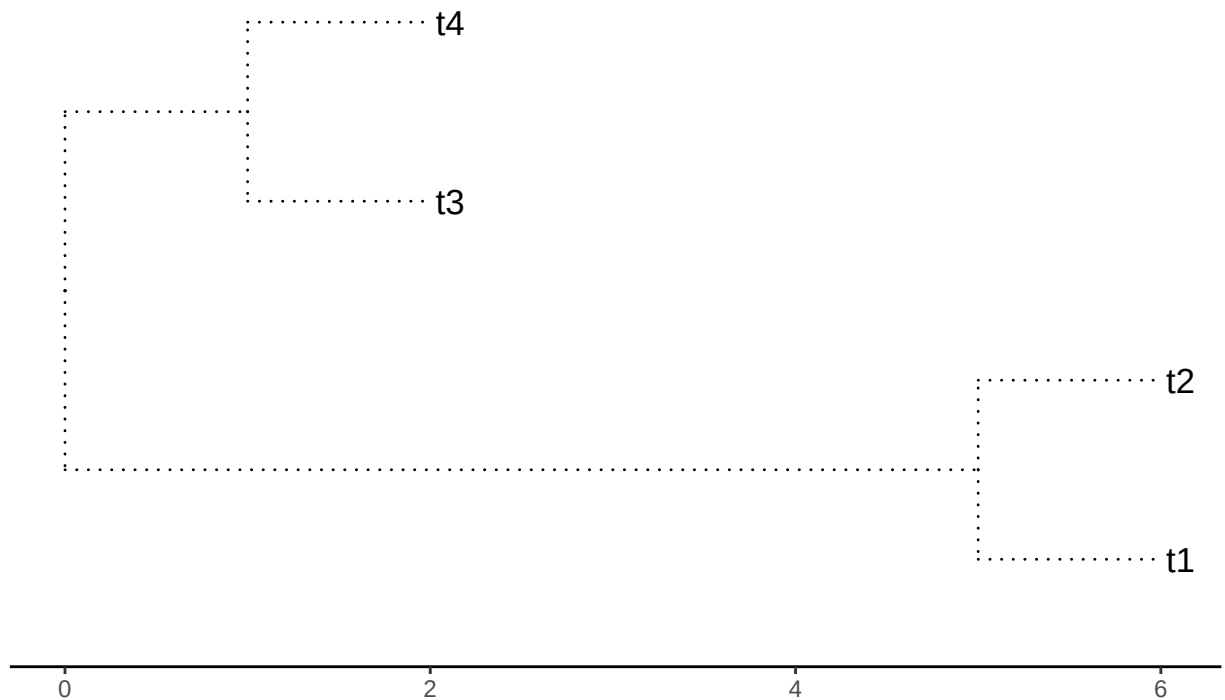
initialTree$edge.length[1] <- 5

plotTree <- ggtree(initialTree, ladderize=TRUE,
                   color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=4.5, color="black") +
  theme_tree2() +
  ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
               ". 2=",initialTree$edge.length[2],".",sep=""))

##
print(plotTree)

```

Four terminals. Branches 1=5. 2=1.



```
for( modelo in c("simpleswap","allswap","uniform") ){
  for( rama in c("terminals","internals","all") ){
    ## cat( "\n\t Branchs swapped=\t",rama,".\t",sep="" )
    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama
                )
    printBlepd(val, compact=TRUE)

    cat( "\n\n" )
  }
}
```

```
## model to test simpleswap reps 100
## AreaSelected Freq
## 1          A1A2  100
##
##
## model to test simpleswap reps 100
## AreaSelected Freq
## 1          A1A2  100
```

```

##
##
## model to test simpleswap reps 100
##   AreaSelected Freq
## 1           A1   14
## 2          A1A2   74
## 3           A2   12
##
##
## model to test allswap reps 100
##   AreaSelected Freq
## 1          A1A2  100
##
##
## model to test allswap reps 100
##   AreaSelected Freq
## 1          A1A2  100
##
##
## model to test allswap reps 100
##   AreaSelected Freq
## 1           A1   30
## 2          A1A2  37
## 3           A2   33
##
##
## model to test uniform reps 100
##   AreaSelected Freq
## 1          A1A2  100
##
##
## model to test uniform reps 100
##   AreaSelected Freq
## 1          A1A2  100
##
##
## model to test uniform reps 100
##   AreaSelected Freq
## 1           A1   54
## 2           A2   46

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)

## data frame with 0 columns and 0 rows
## Now the internals

AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",

```

```

                                approach="all")

printEvalBranch(AllInternals, compact=FALSE)

## data frame with 0 columns and 0 rows
#now with a longer branch

initialTree <- tree

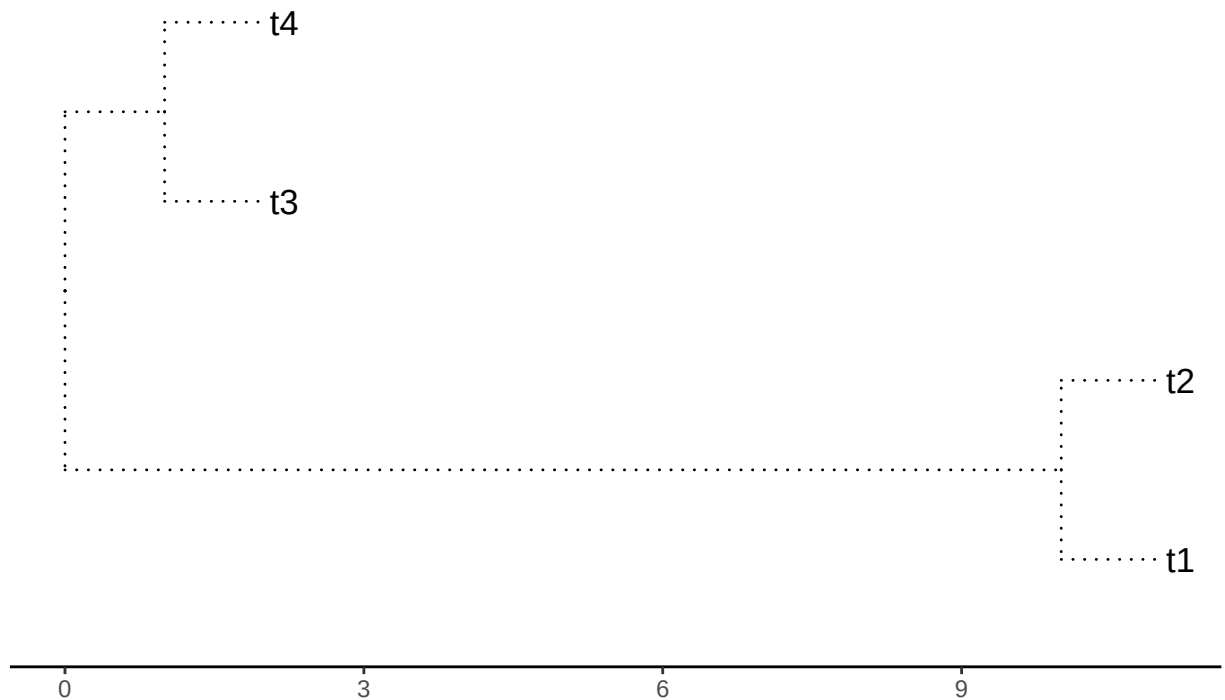
initialTree$edge.length[1] <- 10

plotTree <- ggtree(initialTree, ladderize=TRUE,
                    color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=4.5, color="black") +
  theme_tree2() +
  ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
    ". 2=",initialTree$edge.length[2], ".", sep=""))

##
print(plotTree)

```

Four terminals. Branches 1=10. 2=1.



```

for( modelo in c("simpleswap","allswap","uniform") ){

  for( rama in c("terminals","internals","all") ){

## cat( "\n\t Branchs swapped=\t",rama,".\t",sep="" )
    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama
                  )
    printBlepd(val, compact=TRUE)

    cat( "\n\n" )

  }

}

```

```

## model to test simpleswap reps 100
##   AreaSelected Freq
## 1           A1A2  100
##
##
## model to test simpleswap reps 100
##   AreaSelected Freq
## 1           A1A2  100
##
##
## model to test simpleswap reps 100
##   AreaSelected Freq
## 1           A1    22
## 2           A1A2   70
## 3           A2     8
##
##
## model to test allswap reps 100
##   AreaSelected Freq
## 1           A1A2  100
##
##
## model to test allswap reps 100
##   AreaSelected Freq
## 1           A1A2  100
##
##
## model to test allswap reps 100
##   AreaSelected Freq
## 1           A1    26
## 2           A1A2  42
## 3           A2    32
##
##
## model to test uniform reps 100
##   AreaSelected Freq

```

```

## 1      A1A2  100
##
##
## model to test uniform reps 100
##   AreaSelected Freq
## 1      A1A2  100
##
##
## model to test uniform reps 100
##   AreaSelected Freq
## 1      A1    50
## 2      A2    50

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)

## data frame with 0 columns and 0 rows
## Now the internals

AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")

printEvalBranch(AllInternals, compact=FALSE)

## data frame with 0 columns and 0 rows
##now with a longer branch

initialTree <- tree

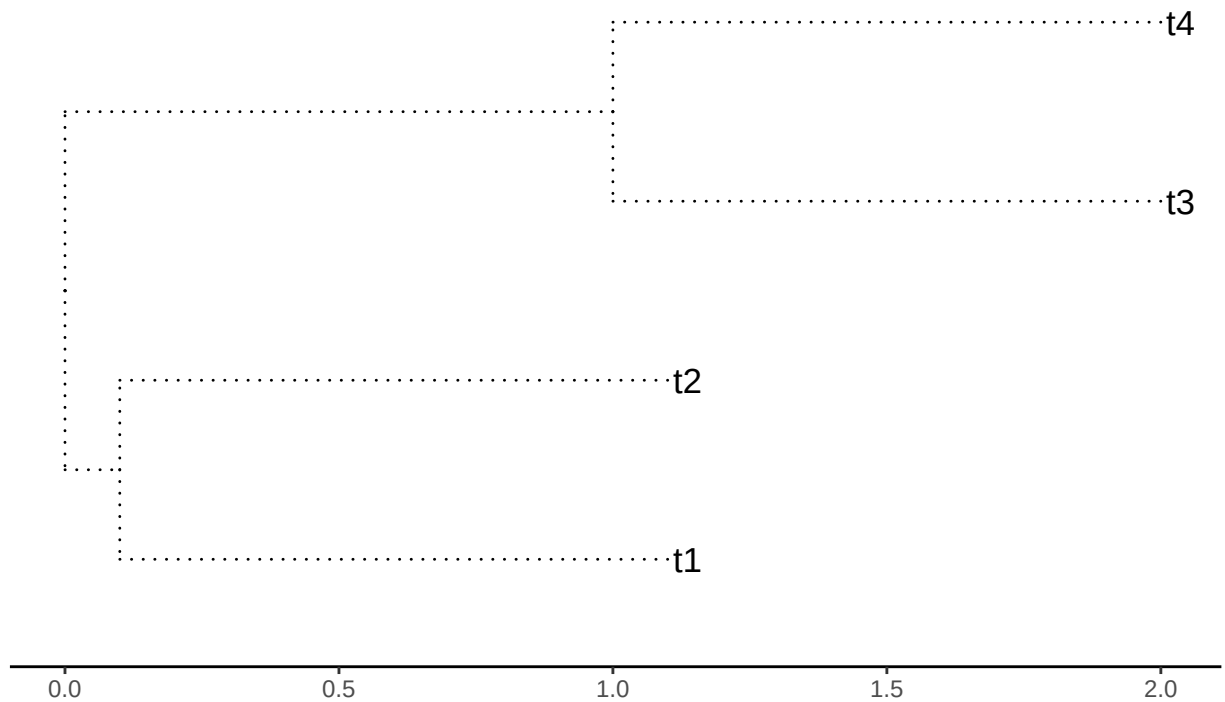
initialTree$edge.length[1] <- 0.1

plotTree <- ggtree(initialTree, ladderize=TRUE,
                   color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=4.5, color="black") +
  theme_tree2() +
  ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
               ". 2=",initialTree$edge.length[2],".",sep=""))

##
print(plotTree)

```

Four terminals. Branches 1=0.1. 2=1.



```
for( modelo in c("simpleswap","allswap","uniform") ){
  for( rama in c("terminals","internals","all") ){
    ## cat( "\n\t Branchs swapped=\t",rama,".\t",sep="" )
    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama
                )
    printBlepd(val, compact=TRUE)

    cat( "\n\n" )
  }
}
```

```
## model to test simpleswap reps 100
## AreaSelected Freq
## 1          A1A2  100
##
##
## model to test simpleswap reps 100
## AreaSelected Freq
## 1          A1A2  100
```

```

##
##
## model to test simpleswap reps 100
##   AreaSelected Freq
## 1           A1   14
## 2          A1A2   72
## 3           A2   14
##
##
## model to test allswap reps 100
##   AreaSelected Freq
## 1          A1A2  100
##
##
## model to test allswap reps 100
##   AreaSelected Freq
## 1          A1A2  100
##
##
## model to test allswap reps 100
##   AreaSelected Freq
## 1           A1   36
## 2          A1A2   35
## 3           A2   29
##
##
## model to test uniform reps 100
##   AreaSelected Freq
## 1          A1A2  100
##
##
## model to test uniform reps 100
##   AreaSelected Freq
## 1          A1A2  100
##
##
## model to test uniform reps 100
##   AreaSelected Freq
## 1           A1   48
## 2           A2   52

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)

## data frame with 0 columns and 0 rows

## Now the internals

AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",

```



```

        approach="all")

printEvalBranch(AllInternals, compact=FALSE)

## data frame with 0 columns and 0 rows
#now with a longer branch in brach 2

#now with a longer branch

initialTree <- tree

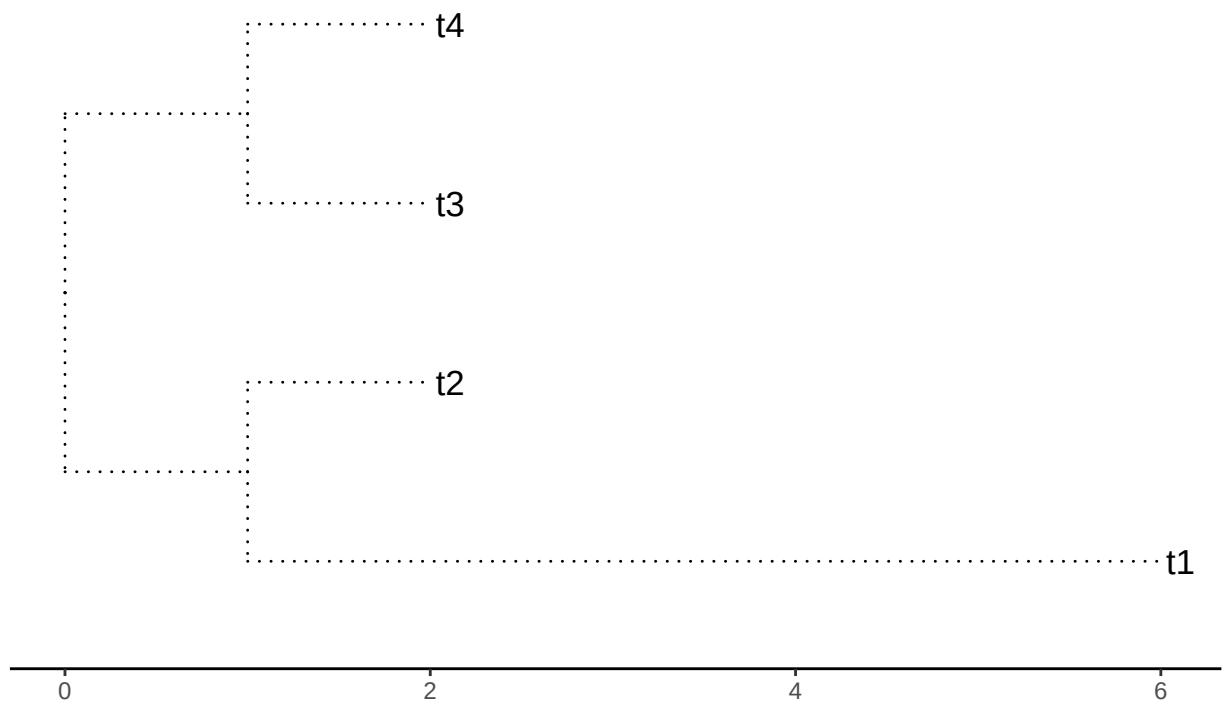
initialTree$edge.length[2] <- 5

plotTree <- ggtree(initialTree, ladderize=TRUE,
                   color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=4.5, color="black") +
  theme_tree2() +
  ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
    ". 2=",initialTree$edge.length[2], ".", sep=""))

##
print(plotTree)

```

Four terminals. Branches 1=1. 2=5.



```
for( modelo in c("simpleswap","allswap","uniform") ){
  for( rama in c("terminals","internals","all") ){
    ## cat( "\n\t Branchs swapped=\t",rama,".\t",sep="" )
    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama
                )
    printBlepd(val, compact=TRUE)

    cat( "\n\n" )
  }
}
```

```
## model to test simpleswap reps 100
## AreaSelected Freq
## 1          A1    65
## 2          A2    35
##
##
## model to test simpleswap reps 100
## AreaSelected Freq
```

```

## 1          A1  100
##
##
## model to test simpleswap reps 100
##   AreaSelected Freq
## 1          A1   65
## 2          A1A2  18
## 3          A2   17
##
##
## model to test allswap reps 100
##   AreaSelected Freq
## 1          A1   51
## 2          A2   49
##
##
## model to test allswap reps 100
##   AreaSelected Freq
## 1          A1  100
##
##
## model to test allswap reps 100
##   AreaSelected Freq
## 1          A1   32
## 2          A1A2  29
## 3          A2   39
##
##
## model to test uniform reps 100
##   AreaSelected Freq
## 1          A1   57
## 2          A2   43
##
##
## model to test uniform reps 100
##   AreaSelected Freq
## 1          A1  100
##
##
## model to test uniform reps 100
##   AreaSelected Freq
## 1          A1   49
## 2          A2   51

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)

## data frame with 0 columns and 0 rows
## Now the internals

```

```

AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")

printEvalBranch(AllInternals, compact=FALSE)

## data frame with 0 columns and 0 rows
#now with a longer branch

initialTree <- tree

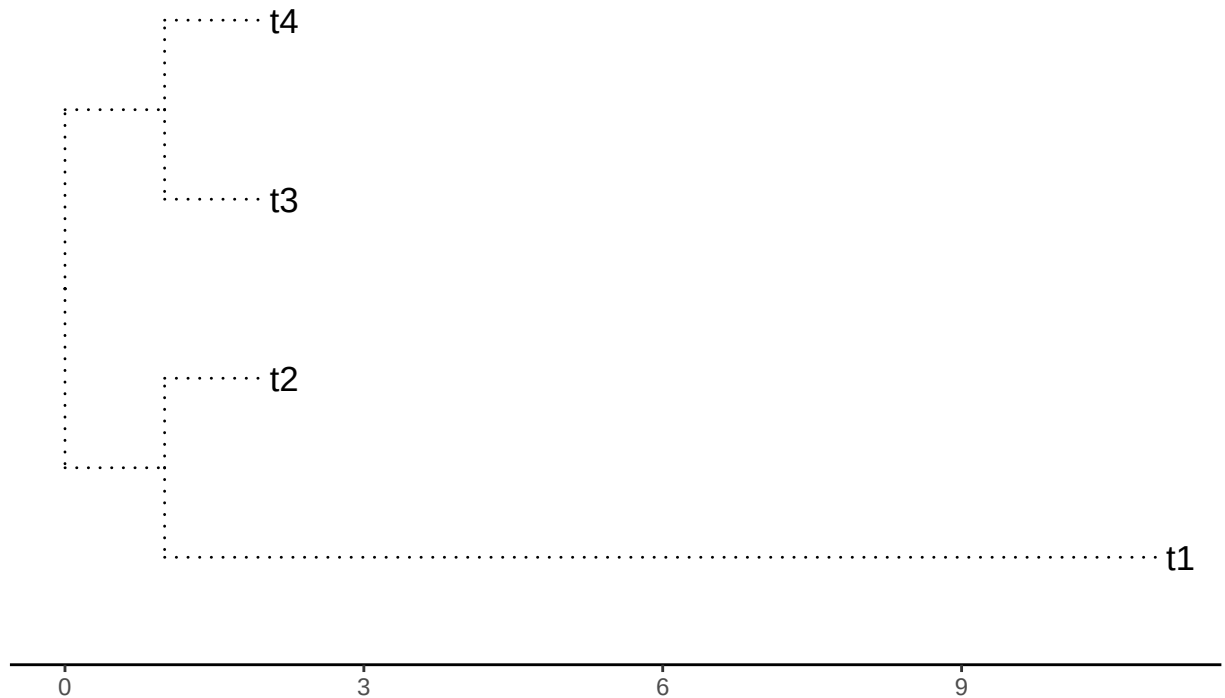
initialTree$edge.length[2] <- 10

plotTree <- ggtree(initialTree, ladderize=TRUE,
                   color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=4.5, color="black") +
  theme_tree2() +
  ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
               ". 2=",initialTree$edge.length[2],".",sep=""))

##
print(plotTree)

```

Four terminals. Branches 1=1. 2=10.



```

for( modelo in c("simpleswap", "allswap", "uniform") ){
  for( rama in c("terminals", "internals", "all") ){
    ## cat( "\n\t Branchs swapped=\t", rama, ".\t", sep="" )
    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama
                )
    printBlepd(val, compact=TRUE)

    cat( "\n\n" )
  }
}

```

```

## model to test simpleswap reps 100
## AreaSelected Freq
## 1          A1    64
## 2          A2    36
##
##
## model to test simpleswap reps 100
## AreaSelected Freq

```

```

## 1          A1  100
##
##
## model to test simpleswap reps 100
##   AreaSelected Freq
## 1          A1   72
## 2          A1A2  16
## 3          A2   12
##
##
## model to test allswap reps 100
##   AreaSelected Freq
## 1          A1   51
## 2          A2   49
##
##
## model to test allswap reps 100
##   AreaSelected Freq
## 1          A1  100
##
##
## model to test allswap reps 100
##   AreaSelected Freq
## 1          A1   29
## 2          A1A2  28
## 3          A2   43
##
##
## model to test uniform reps 100
##   AreaSelected Freq
## 1          A1   48
## 2          A2   52
##
##
## model to test uniform reps 100
##   AreaSelected Freq
## 1          A1  100
##
##
## model to test uniform reps 100
##   AreaSelected Freq
## 1          A1   49
## 2          A2   51

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)

## data frame with 0 columns and 0 rows
## Now the internals

```

```

AllInternals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")

printEvalBranch(AllInternals, compact=FALSE)

## data frame with 0 columns and 0 rows
#now with a longer branch

initialTree <- tree

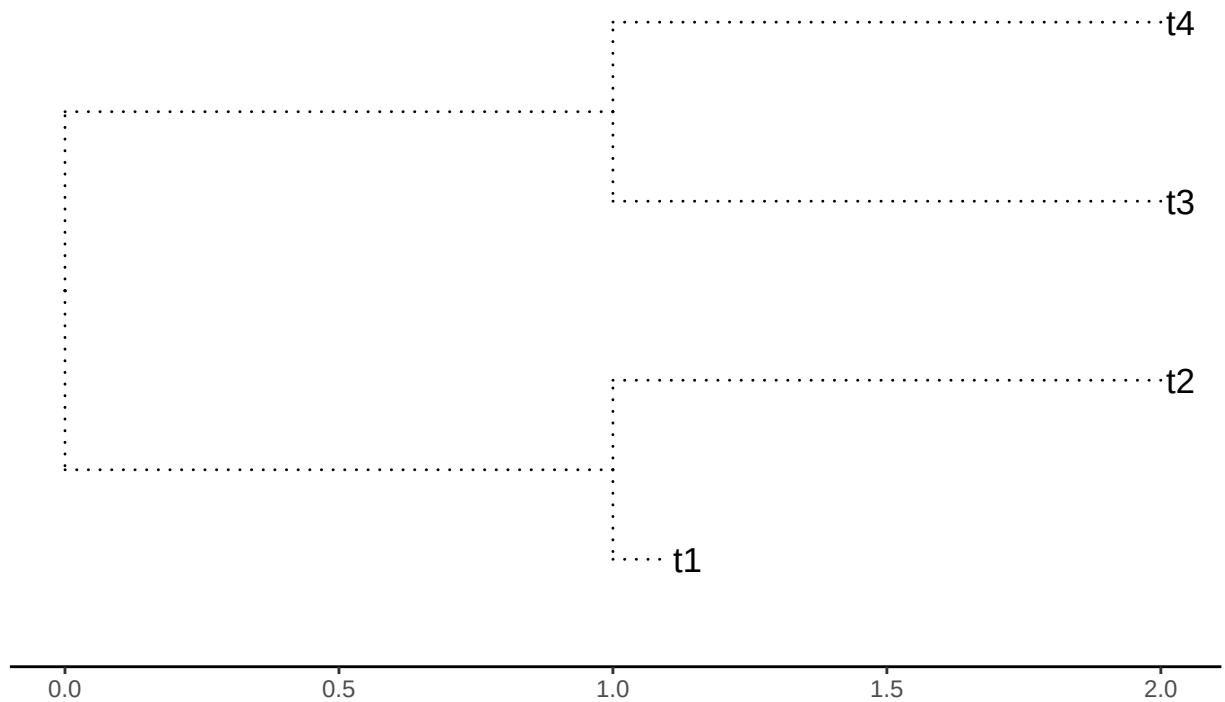
initialTree$edge.length[2] <- 0.1

plotTree <- ggtree(initialTree, ladderize=TRUE,
                   color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=4.5, color="black") +
  theme_tree2() +
  ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
               ". 2=",initialTree$edge.length[2], ".", sep=""))

##
print(plotTree)

```

Four terminals. Branches 1=1. 2=0.1.



```
for( modelo in c("simpleswap","allswap","uniform") ){
  for( rama in c("terminals","internals","all") ){
    ## cat( "\n\t Branchs swapped=\t",rama,".\t",sep="" )
    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama
                )
    printBlepd(val, compact=TRUE)

    cat( "\n\n" )
  }
}
```

```
## model to test simpleswap reps 100
## AreaSelected Freq
## 1          A1    34
## 2          A2    66
##
##
## model to test simpleswap reps 100
## AreaSelected Freq
```



```

## 1          A2  100
##
##
## model to test simpleswap reps 100
##   AreaSelected Freq
## 1          A1    8
## 2          A1A2  15
## 3          A2   77
##
##
## model to test allswap reps 100
##   AreaSelected Freq
## 1          A1   55
## 2          A2   45
##
##
## model to test allswap reps 100
##   AreaSelected Freq
## 1          A2  100
##
##
## model to test allswap reps 100
##   AreaSelected Freq
## 1          A1   29
## 2          A1A2  45
## 3          A2   26
##
##
## model to test uniform reps 100
##   AreaSelected Freq
## 1          A1   46
## 2          A2   54
##
##
## model to test uniform reps 100
##   AreaSelected Freq
## 1          A2  100
##
##
## model to test uniform reps 100
##   AreaSelected Freq
## 1          A1   47
## 2          A2   53

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)

## data frame with 0 columns and 0 rows
## Now the internals

```

```
AllInternals <- evalBranch(tree=initialTree,  
                           distribution = dist4taxa,  
                           branchToEval="internals",  
                           approach="all")  
  
printEvalBranch(AllInternals, compact=FALSE)  
  
## data frame with 0 columns and 0 rows
```